CSCI 4022 - Final Project Report

An Exploration of Wikipedia's Best Ranked Pages Using PageRank and MinHash Analysis

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1 Introduction

Wikipedia is one of the largest publicly available collections of connected information on the internet. Although users usually experience Wikipedia as individual article pages, the site can also be studied as a large directed graph where articles are nodes and hyperlinks are edges [4]. In this project, we analyze English Wikipedia using this graph structure to investigate how structural importance relates to article similarity. We compute PageRank to identify highly connected and influential pages, then compare the text similarity of top-ranked pages against lower-ranked, random, and category-based groups using exact Jaccard similarity and MinHash [1, 2, 5]. The goal is to determine whether pages that are important in Wikipedia's link network are also more semantically diverse than other groups of pages. Our implementation uses Python with pandas, NumPy, DuckDB, PyArrow, Matplotlib, and PowerShell, and the full project code is available at our GitHub repository. Here is a link to our presentation video.

2 Research Questions

This project investigates how Wikipedia's graph structure relates to the semantic similarity of its articles. PageRank gives us a way to identify articles that are structurally important within the hyperlink network, while Jaccard similarity and MinHash allow us to compare the textual overlap between selected groups of articles [1, 2, 5]. Using these methods, we focus on the following research questions:

1. Does selecting top-ranked pages by PageRank produce a set of documents that is more semantically diverse than selecting pages by category or random sampling?
2. Are highly ranked pages more or less similar to each other than lower-ranked pages?
3. How does article similarity vary across PageRank percentile ranges?

Our initial hypothesis, outlined in our project proposal, was that top PageRank pages would be more semantically diverse because highly ranked pages often serve as broad connector pages across Wikipedia. In contrast, we expected lower-ranked pages to be more niche and therefore more internally similar. To test this, we compared the average pairwise Jaccard similarity within top PageRank, median PageRank, bottom PageRank, random, and category-based groups. Lower average similarity was interpreted as greater semantic diversity, while higher average similarity indicated that the pages in a group shared more phrase-level content.

3 Methods

3.1 Data Description

This project used publicly available English Wikipedia dump files from Wikimedia [3]. The graph structure was constructed from SQL dump files containing page metadata, link targets, page-to-page links, and category links. Article text was extracted separately from the compressed Wikipedia pages-articles XML dump. Together, these files allowed us to represent Wikipedia as both a directed graph and a collection of article documents for text similarity analysis.

The data was taken from Wikimedia's English Wikipedia "latest" dump directory, meaning the files correspond to the most recent English Wikipedia dump available at the time they were downloaded rather than a fixed historical dump folder [3]. We pulled all data on April 27, 2026. Therefore, this project analyzes the English Wikipedia dump that was current as of approximately April 27, 2026.

The raw dump files were first converted into Parquet format to make later processing more efficient. This step reduced the size and complexity of the original SQL files by keeping only the columns needed for graph construction and analysis. From the page metadata, we kept each page's id, namespace, title, redirect status, and length. From the link data, we kept the source page id and target link id, which were later joined with the link target table to recover target article titles.

After conversion, the page table was cleaned so that only standard Wikipedia article pages were retained. Specifically, we kept pages in namespace 0 (keeping only main Wikipedia articles, excluding talk pages, templates, files, categories, and user pages), removed redirect pages, and filtered out extremely short pages. This produced the final node set for the graph. The cleaned graph contained 7,160,764 article nodes.

The directed edge list was built from the Wikipedia internal hyperlink data. Each edge represents a hyperlink from one article to another article, matching the common representation of Wikipedia link networks as directed graphs [4]. To construct these edges, page links were joined with the link target table and then matched against the cleaned article page table. Edges were kept only when both the source and target were valid article pages. Self-links were removed, and repeated links between the same pair of articles were deduplicated so that each directed article-to-article connection appeared only once. The final graph contained 642,411,155 directed edges.

For the text similarity portion of the project, article text was extracted from the Wikipedia XML dump. The extraction process removed common wiki markup, references, tables, templates, file links, category links, HTML tags, and excess whitespace. The cleaned article text was then joined with the cleaned page table so that similarity analysis was only performed on pages that were also part of the graph. Articles with fewer than 100 words or missing cleaned text were removed. The resulting clean article-text dataset contained 3,302,036 usable articles.

3.2 Algorithms, Implementation, and Software

The main graph algorithm used in this project was PageRank [1]. PageRank assigns each page a structural importance score based on the number and quality of incoming links it receives. Intuitively, a page receives a higher PageRank score when it is linked to by many other pages, especially if those linking pages are themselves important.

PageRank was computed using power iteration [1]. The update rule used in our implementation was

$$\mathbf{r}^{(t+1)} = \beta A \mathbf{r}^{(t)} + \frac{1-\beta}{N}\mathbf{1} + \beta \frac{\sum_{j \in D} r_j^{(t)}}{N}\mathbf{1}$$

where $\mathbf{r}^{(t)}$ is the PageRank vector at iteration t, β is the damping factor, A is the column-stochastic adjacency/transition matrix, N is the number of pages, $\mathbf{1}$ is a vector of ones, and D is the set of dangling nodes with no outgoing links. The first term represents PageRank mass passed through outgoing links, the second term represents uniform teleportation, and the final term redistributes PageRank mass from dangling nodes uniformly across all pages.

The algorithm initialized every article with equal PageRank mass. At each iteration, each page distributed its current PageRank score across its outgoing links. A damping factor of $\beta = 0.85$ was used, meaning that the random surfer followed links with probability 0.85 and teleported uniformly to any page with probability $1 - \beta = 0.15$ [1]. Dangling nodes, or pages with no outgoing links, were handled by redistributing their PageRank mass uniformly across the graph during each iteration.

The PageRank computation used a convergence tolerance of 10^{-8} and a maximum of 100 iterations. The final PageRank output included each page's id, title, PageRank score, rank, percentile, in-degree, and out-degree.

After PageRank was computed, pages were selected into experimental groups for text similarity analysis. The primary groups were the top PageRank pages, median PageRank pages, bottom PageRank pages, and a random sample of pages. Additional category-based groups were also created for topics such as biology, computer science, history, mathematics, philosophy, and physics. Only pages with usable cleaned article text were included in these groups.

To compare article similarity, each selected article was transformed into a set of word shingles [5]. A shingle is a sequence of consecutive words from a document. This project used 3-word shingles, meaning that each article was represented as a set of overlapping three-word phrases. For example, the phrase "graph analysis of Wikipedia pages" would produce shingles such as "graph analysis of," "analysis of Wikipedia," and "of Wikipedia pages." This representation preserves more local context than individual words while still allowing efficient set-based comparison.

Exact Jaccard similarity was used to measure pairwise document similarity [2, 5]. For two articles represented by shingle sets A and B, exact Jaccard similarity is defined as

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}.$$

A Jaccard similarity score close to 0 means that two articles share very little phrase-level content, while a score closer to 1 means that the articles have substantial overlap. For this project, lower average pairwise similarity within a group was interpreted as greater semantic diversity.

MinHash was used as an approximate method for estimating Jaccard similarity [5]. Each article's shingle set was converted into a MinHash signature using 128 hash permutations. The MinHash estimate between two articles was calculated as the fraction of matching positions in their signatures. This allowed us to evaluate whether MinHash provided a reasonable approximation of exact Jaccard similarity on the selected article groups.

The main software tools used for the project were Python, pandas, NumPy, DuckDB, PyArrow, Matplotlib, and PowerShell. Python was used to implement the full data processing and analysis pipeline. Pandas and PyArrow were used for tabular data processing and Parquet storage. DuckDB was used for large joins during graph construction. NumPy was used for the PageRank power iteration implementation, and Matplotlib was used to generate the final figures for the report. PowerShell scripts were used to download the raw data dumps, validate them, and orchestrate the end-to-end Python pipeline.

4 Results

This section presents the main outputs of the PageRank and similarity analysis. The results are organized around the major parts of the project: identifying the best ranked Wikipedia pages, examining the PageRank distribution, comparing similarity across experimental groups, validating MinHash against exact Jaccard similarity, and interpreting how similarity changes across PageRank percentile ranges.

4.1 Best Ranked Pages

The PageRank algorithm produced a structural importance score for each article in the cleaned Wikipedia graph. The highest-ranked pages were articles that received many incoming links from other pages, especially from pages that were

themselves well-connected. The top-ranked articles in the current run included `Geographic_coordinate_system`, `Wayback_Machine`, `United_States`, `Wikidata`, `Time_zone`, and `Taxonomy_(biology)`.

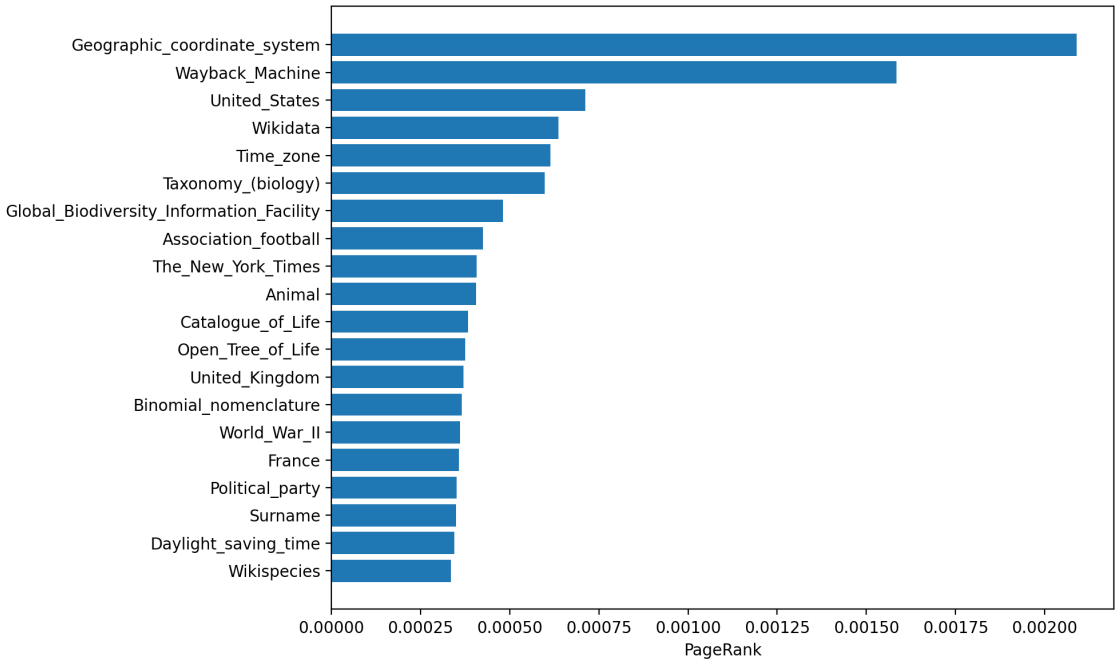

Figure 1: Top 20 Wikipedia articles by PageRank score.

Figure 1 acts as a sanity check for the PageRank implementation. The highest-ranked pages are not random isolated topics; they are broad reference pages that are frequently linked across many areas of Wikipedia. For example, pages related to geographic coordinates, archived references, countries, time zones, taxonomy, and major knowledge databases appear near the top. This supports the idea that PageRank is capturing structural centrality in the hyperlink network rather than simply ranking pages by article length or topic popularity.

4.2 PageRank Distribution

The PageRank scores were highly skewed. Most Wikipedia articles had extremely small PageRank values, while a small number of pages had much larger scores. This is expected for a large hyperlink graph because links are not evenly distributed across all pages. A small set of articles receive a very large number of incoming links, while most articles receive relatively few.

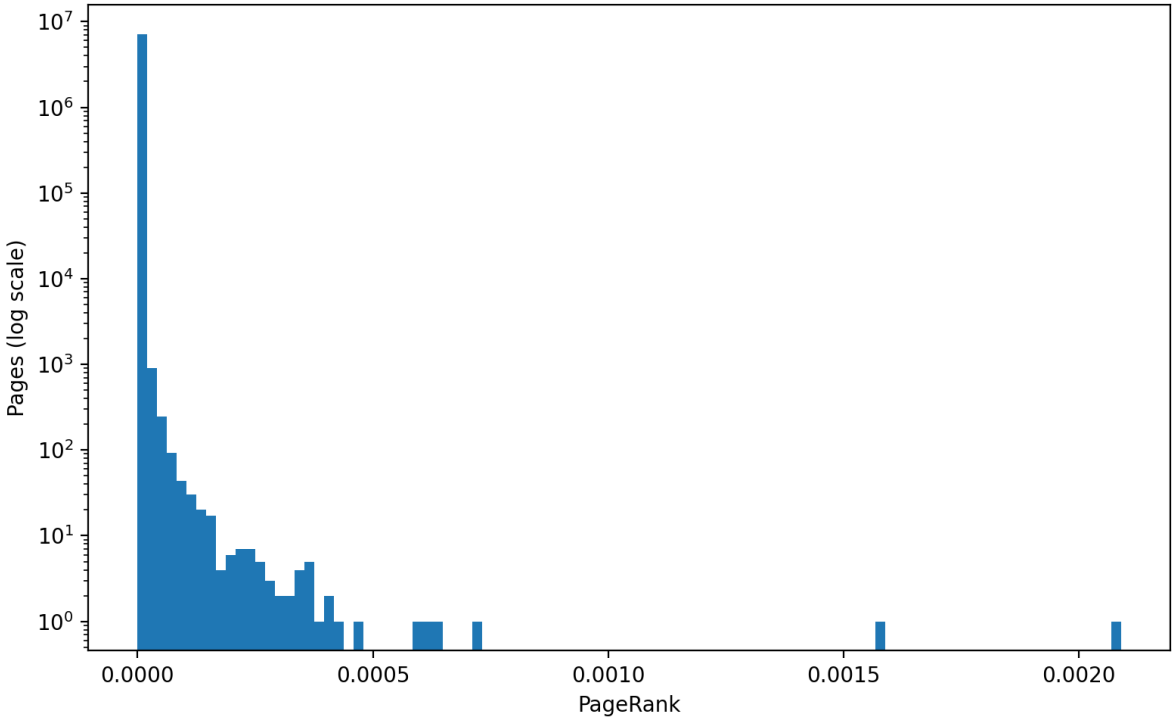

Figure 2: Distribution of PageRank scores across the cleaned Wikipedia article graph. The y-axis is shown on a logarithmic scale.

Figure 2 shows this heavy-tailed pattern. The logarithmic y-axis makes it easier to see that the vast majority of articles are concentrated near very low PageRank values. Only a small number of pages appear in the far right tail of the distribution. This result is important because it confirms that Wikipedia has a strongly unequal link structure: a relatively small set of pages function as major hubs or reference points in the article network.

4.3 Similarity by Group

After computing PageRank, we selected article groups based on rank position and compared their internal text similarity. The main groups were `top_pagerank`, `median_pagerank`, `bottom_pagerank`, and `random`. Category-based groups were also included for comparison, though their sample sizes were much smaller and are therefore interpreted more cautiously.

Each article was represented as a set of 3-word shingles, and exact Jaccard similarity was computed for every pair of articles within each group. Lower mean similarity indicates greater semantic diversity because the articles share less phrase-level content.

Group	Pages	Pairs	Mean Similarity	Median Similarity
`top_pagerank`	500	124750	0.004164	0.003248
`bottom_pagerank`	500	124750	0.001269	0.000000
`random`	500	124750	0.000769	0.000000
`median_pagerank`	500	124750	0.000700	0.000000
`category_history`	8	28	0.004617	0.004345
`category_mathematics`	2	1	0.004313	0.004313
`category_philosophy`	4	6	0.003725	0.001081
`category_computer_science`	7	21	0.002507	0.000923
`category_biology`	10	45	0.001483	0.001135
`category_physics`	9	36	0.001390	0.000748

Table 1: Average exact Jaccard similarity by experimental group.

The most important result is that the `top_pagerank` group had a higher mean similarity than the random, median PageRank, and bottom PageRank groups. This does not support our original hypothesis. We expected top PageRank pages to be more semantically diverse because they are broadly connected across Wikipedia. Instead, the top PageRank group had a mean exact Jaccard similarity of 0.004164, compared to 0.000769 for the random group, 0.000700 for the median PageRank group, and 0.001269 for the bottom PageRank group.

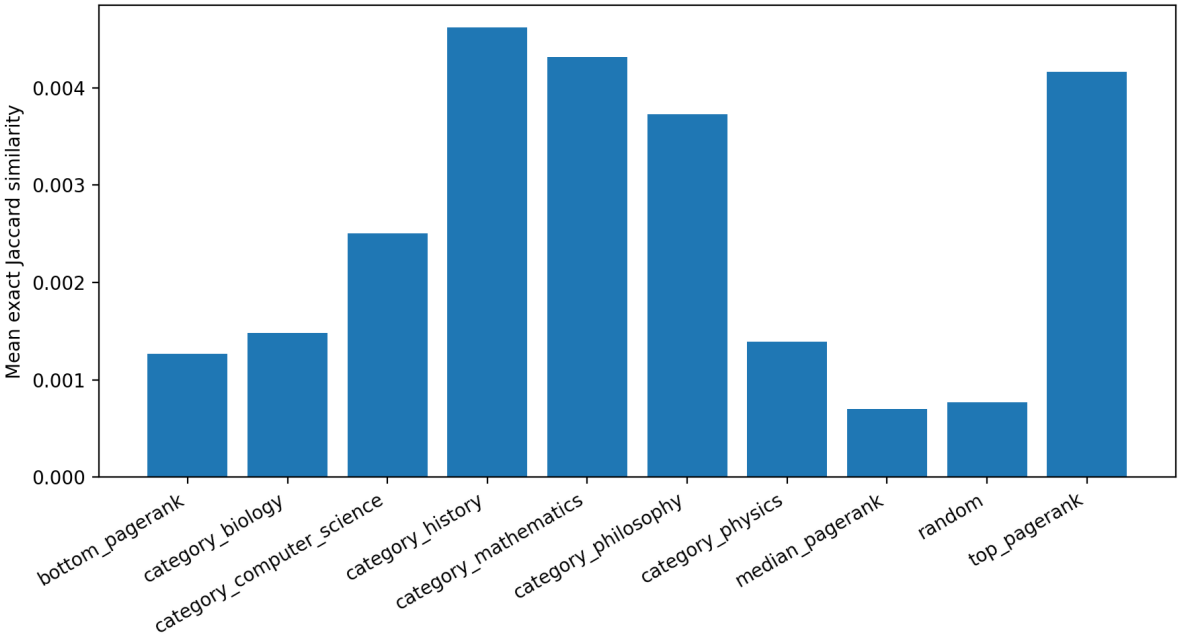

Figure 3: Mean exact Jaccard similarity by group. Lower values indicate greater semantic diversity.

Figure 3 shows that the top PageRank group is one of the more internally similar groups. The category groups for history, mathematics, and philosophy also show relatively high mean similarity, but these groups contain very few pages, so they should not be compared as strongly as the 500-page PageRank and random groups.

One possible explanation for the higher similarity among top PageRank pages is that many of these pages are broad reference or infrastructure topics. Pages about geographic coordinates, time zones, countries, taxonomy, databases, and reference systems may share repeated encyclopedic language, common formatting patterns, or similar contextual vocabulary. As a result, these pages can be structurally central while still having overlapping text features under a 3-word shingle similarity metric.

4.4 Pairwise Similarity Spread

The mean similarity values alone do not show whether the results are caused by many small differences or by a few extreme outlier pairs. To examine this, we also plotted the full distribution of pairwise similarities within each group.

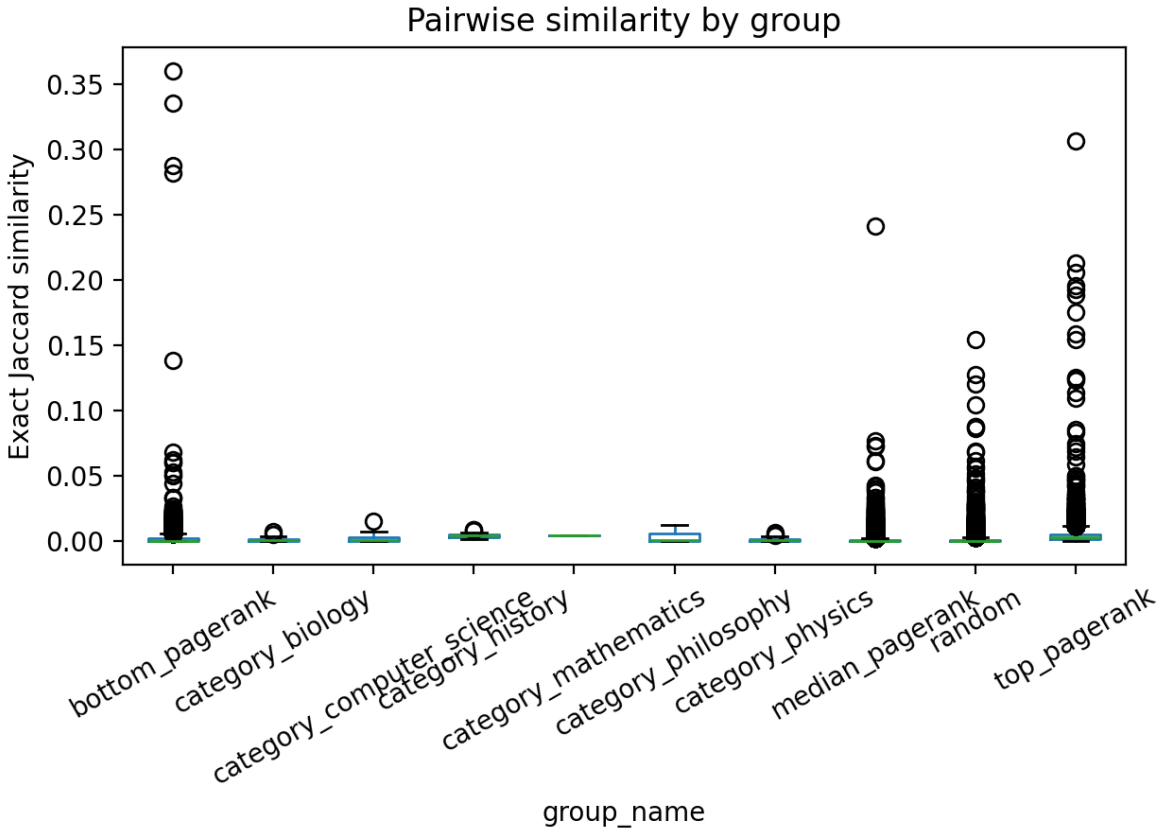

Figure 4: Distribution of pairwise exact Jaccard similarity values within each experimental group.

Figure 4 shows that most pairwise similarities are close to zero across all groups. This is expected because Wikipedia articles usually have distinct text even when they are related. However, the top PageRank and bottom PageRank groups contain more high-similarity outliers than the random and median groups. The top PageRank group in particular has several article pairs with noticeably higher overlap. This helps explain why its average similarity is higher than the random and median PageRank baselines.

The boxplot also reinforces that the category groups should be treated carefully. Since some category groups contain only a handful of pages, their distributions are based on very few pairwise comparisons. These values are useful as exploratory results, but they should not be treated as equally reliable as the larger 500-page groups.

4.5 Similarity Across PageRank Percentiles

To examine whether similarity changes gradually across PageRank levels, we also compared mean exact Jaccard similarity across PageRank percentile bins. This analysis divides pages into percentile ranges from `0-10%` through `90-100%` and computes average pairwise similarity within each bin.

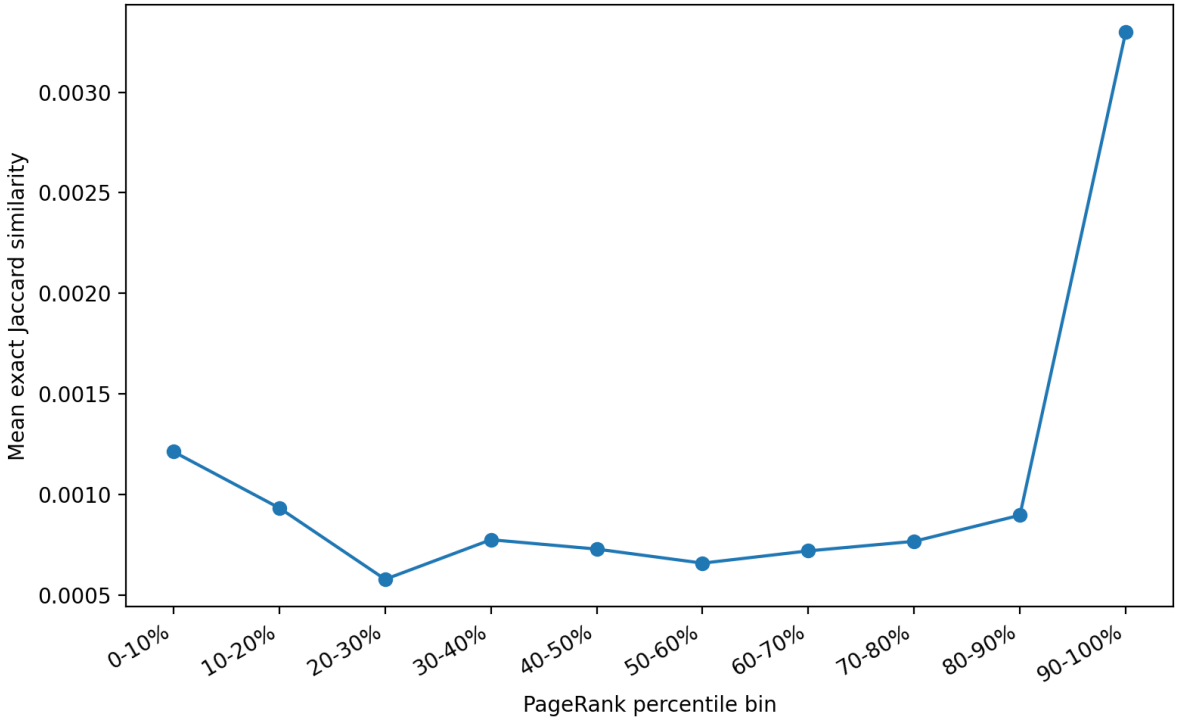

Figure 5: Mean exact Jaccard similarity across PageRank percentile bins.

Figure 5 shows that the highest PageRank percentile bin has the largest mean similarity. The `90-100%` bin has a mean similarity of approximately 0.003299, while most middle percentile bins are below 0.001. This pattern is consistent with the group-level result: articles with the highest PageRank scores are not the most semantically diverse under our similarity measure. Instead, they appear to have greater phrase-level overlap than many lower-ranked or middle-ranked articles.

At the same time, this percentile result should be interpreted cautiously because the percentile bins were not evenly sampled. Some bins contain many more selected pages than others. Therefore, the percentile plot is best understood as an exploratory view of the trend rather than a final statistical test.

4.6 MinHash Validation

Because the project proposal included MinHash as an approximate method for estimating Jaccard similarity [5], we compared MinHash estimates against exact Jaccard values. This validation checks whether MinHash captured the same general similarity patterns as the exact computation.

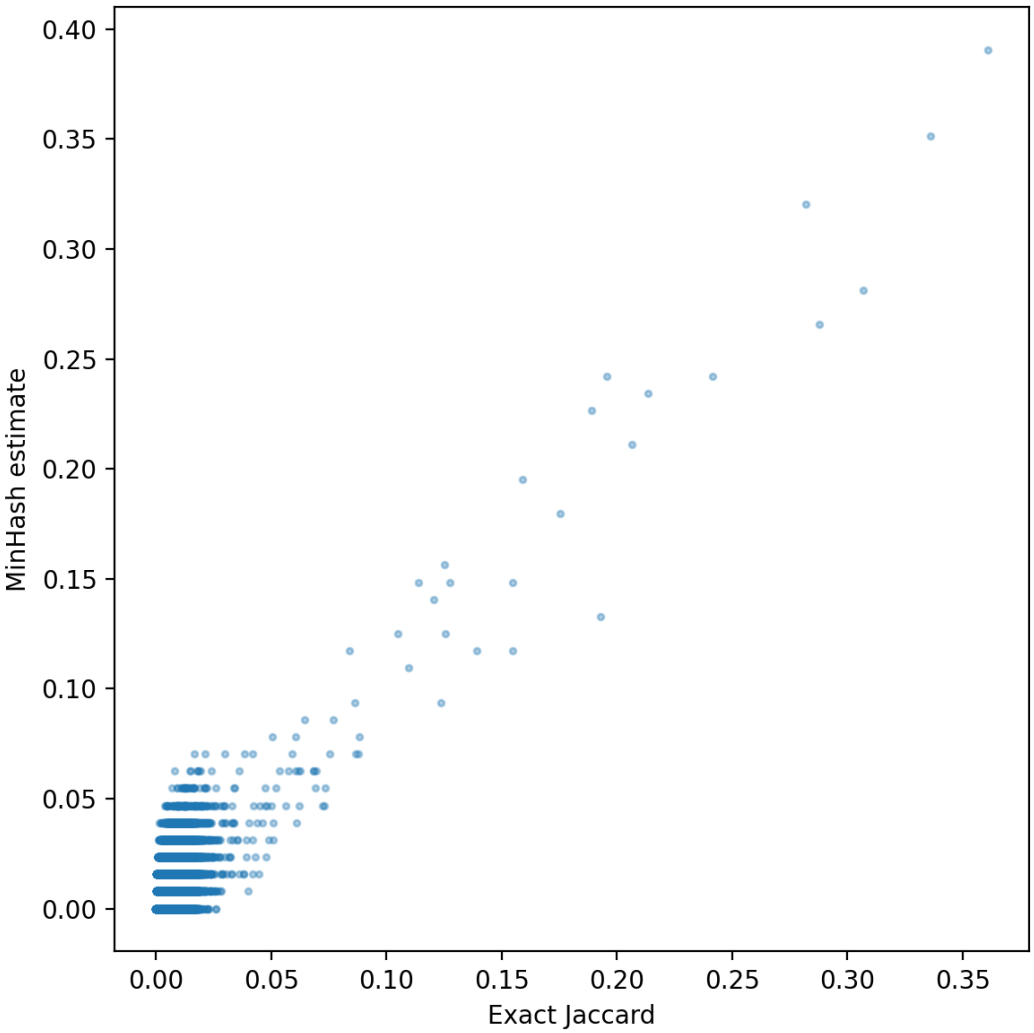

Figure 6: Comparison of MinHash estimated similarity against exact Jaccard similarity.

Figure 6 shows a positive relationship between exact Jaccard similarity and MinHash estimated similarity. Pairs with higher exact Jaccard values generally also have higher MinHash estimates. The mean absolute error between MinHash and exact Jaccard similarity was approximately 0.002187. This suggests that MinHash was a reasonable approximation for identifying broad similarity patterns, although exact Jaccard was still used for the main reported group comparisons.

5 Conclusion

5.1 Findings

Overall, the results show that PageRank centrality is related to article similarity, but not in the direction we originally expected. The original hypothesis predicted that top-ranked pages would be more semantically diverse because they are broadly connected across Wikipedia. However, the current results show that top PageRank pages had higher average similarity than random, median-ranked, and bottom-ranked pages.

The main findings are:

1. The Wikipedia PageRank distribution is highly skewed, with most articles receiving very small PageRank scores and a small number of articles receiving much larger scores.
2. The top PageRank pages are broad, highly referenced articles such as `Geographic_coordinate_system`, `Wayback_Machine`, `United_States`, `Wikidata`, and `Time_zone`.
3. The `top_pagerank` group had higher mean exact Jaccard similarity than the random, median PageRank, and bottom PageRank groups.
4. The highest PageRank percentile bin also had the highest mean similarity among the percentile bins.
5. MinHash approximated exact Jaccard similarity reasonably well for the selected document pairs, with a mean absolute error of approximately 0.002187.

These findings suggest that structurally central Wikipedia pages may share more text-level features than expected. Rather than being the most semantically diverse pages, the highest PageRank articles may form a collection of broad reference topics that use overlapping encyclopedic language and are frequently linked throughout Wikipedia.

5.2 Limitations and Future Work

There are several limitations to this project that affect how the results should be interpreted. First, the similarity metric used in this analysis is based on exact Jaccard similarity over 3-word shingles [2, 5]. This captures phrase-level lexical overlap, but it does not fully measure semantic meaning. Two articles could discuss similar concepts using different language and receive a low Jaccard similarity score, while two articles with repeated structural or encyclopedic phrasing could receive a higher score even if their topics are not deeply similar.

Second, the text cleaning process was intentionally lightweight. The cleaner removed many common forms of wiki markup, including references, tables, templates, file links, category links, HTML tags, and excess whitespace. However, it was not a complete MediaWiki parser. Some markup remnants, boilerplate text, or repeated formatting patterns may remain in the cleaned article text. These artifacts could increase measured similarity between pages, especially for broad reference articles that share common Wikipedia-style language.

Third, the category-based groups were much smaller than the PageRank and random groups. While the top PageRank, median PageRank, bottom PageRank, and random groups each contained 500 pages, several category groups contained fewer than 10 pages. This happened because the category selection process matched pages directly assigned to broad category names, rather than expanding through subcategories. As a result, the category results should be treated as exploratory and should not be given the same weight as the larger PageRank-based comparisons.

Fourth, the percentile-bin analysis was not evenly sampled across all PageRank percentile ranges. Some bins contained many more selected pages than others, which means the percentile plot should be interpreted as a descriptive trend rather than a balanced statistical comparison. A stronger future version of this analysis would sample the same number of pages from each percentile bin before computing pairwise similarity.

Finally, PageRank measures structural importance in the hyperlink network, not semantic importance or conceptual breadth directly. A page can receive a high PageRank score because it is frequently used as a reference, appears

in common templates, or is linked by many articles, even if it is not necessarily more conceptually diverse. This distinction is important because our results suggest that structural centrality and semantic diversity are not the same property.

Future work could improve this project by expanding category groups through subcategory traversal, using a more advanced Wikipedia parser, applying stopword or boilerplate removal, and comparing Jaccard similarity with embedding-based semantic similarity methods. A future version could also run balanced samples across every PageRank percentile bin and use bootstrap confidence intervals or permutation tests to determine whether the observed group differences are statistically reliable.

6 References

References

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